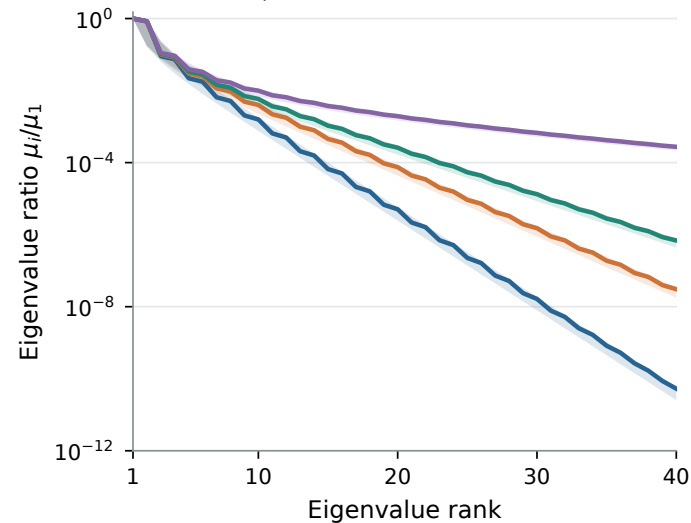


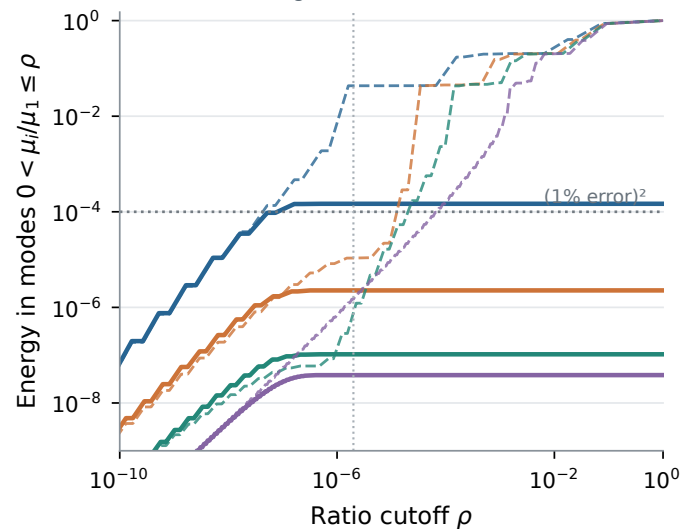
A Gamma sets relative learning rates

Lines: finite spectrum Bands: theorem intervals



B Adam leaves slow-mode residual

Dashed: initial target Solid: Adam at 200k



C The theorem brackets GD delay

Adam: first loss-EMA crossing, half-life 100

